Myungsin Cho

Research Statement

My research lies at the intersection of stable homotopy theory and arithmetic, with an emphasis on equivariant phenomena. My central goal is to understand duality and descent in Real algebraic K-theory (KR). This theory incorporates an involution into algebraic K-theory: forgetting its C_2-symmetry recovers ordinary K-theory, while taking fixed points gives Grothendieck–Witt theory (GW), which studies modules equipped with nondegenerate bilinear forms. I pursue this goal through arithmetic duality, Real trace methods, and the algebraic structures intrinsic to genuine equivariance.

Arithmetic duality and Real algebraic K-theory

Tate–Poitou duality relates global arithmetic information to local data through *étale cohomology*. Thomason descent connects this cohomology to $K(1)$-local algebraic K-theory, suggesting a corresponding duality at the homotopical level. Blumberg and Mandell established this global duality at odd primes. The prime 2 remained open because real embeddings introduce additional cohomological behavior. My dissertation resolves this case. For a number field F, the completion cofiber sequence

$$L_{K(1)}\,\mathrm{K}(\mathcal{O}_F[\tfrac{1}{2}]) \xrightarrow{\kappa} \prod_{\nu|2} L_{K(1)}\,\mathrm{K}(F_\nu) \longrightarrow \mathrm{Cof}(\kappa)$$

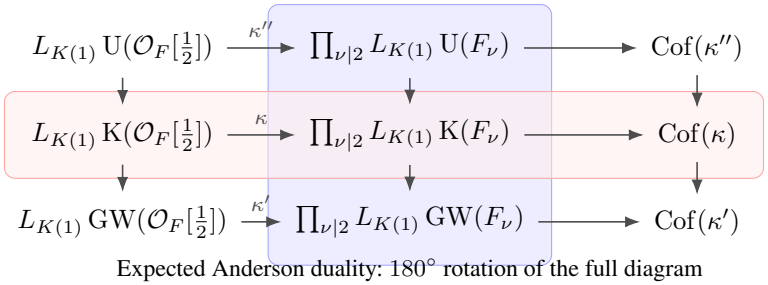
admits a canonical 2-adic Anderson self-duality: the end terms are dual to one another, and the middle term is self-dual. This realizes arithmetic duality as a structural relationship among spectra, rather than only among their homotopy groups.

The theorem also identifies the fiber of the cyclotomic trace for the sphere. For $F = \mathbb{Q}$, known comparison results give, in nonnegative degrees,

$$I_{\mathbb{Z}_2} L_{K(1)}\,\mathrm{K}(\mathbb{Z}) \simeq \Sigma\,\mathrm{Fib}\big(\mathrm{K}(\mathbb{S})^\wedge_2 \xrightarrow{\mathrm{trc}} \mathrm{TC}(\mathbb{S})^\wedge_2\big).$$

Thus arithmetic information in $K(\mathbb{Z})$ determines the part of $K(\mathbb{S})$ not detected by the cyclotomic trace.

In current work with Lucy Yang, we establish the fixed-point counterpart over 2-adic local fields: Anderson duality interchanges the end terms of the hyperbolic sequence $L_{K(1)}\,\mathrm{U} \to L_{K(1)}\,\mathrm{K} \to L_{K(1)}\,\mathrm{GW}$ and preserves its middle term. Our ongoing work seeks to globalize this symmetry through the following diagram, whose rows are completion sequences and whose columns are hyperbolic sequences. The highlighted row and column record the global K-theoretic and local GW dualities already obtained.



Expected Anderson duality: 180° rotation of the full diagram

Toward equivariant duality and descent

The expected rotational symmetry points toward a single duality in genuine C_2-spectra. For a number field with trivial involution, I conjecture

$$I_{\mathbb{Z}_2}^{C_2} L_{K(1)} \mathrm{KR}(\mathcal{O}_F[\tfrac{1}{2}]) \simeq \mathrm{Cof}\big(\kappa_{\mathrm{KR}}^{[-1]}\big),$$

where $\kappa_{\mathrm{KR}}^{[-1]}$ is the completion map for KR with the duality on perfect modules shifted by $[-1]$. Forgetting the C_2-action recovers my dissertation theorem. In ordinary K-theory, arithmetic duality descends from cohomology to spectra. Since a corresponding equivariant cohomology theory for KR is not yet available, I use the expected duality to guide the construction of a Real descent theory.

Real trace methods

Trace methods approximate algebraic K-theory by more computable invariants and also enter arithmetic duality. With Gerhardt, Keenan, Moreno, and Quigley, I develop homological Real trace methods refining the methods of Bruner and Rognes. We construct a multiplicative spectral sequence from Real topological Hochschild homology to the continuous homology of Real topological cyclic homology TCR^-, compatible with equivariant power operations. We determine its differentials and E_∞-pages for $MU_{\mathbb{R}}$, $BP_{\mathbb{R}}$, and several truncated Brown–Peterson theories. My next goal is a Real Segal conjecture for $MU_{\mathbb{R}}$ and $BP_{\mathbb{R}}$, providing further computational input for investigating duality beyond number fields.

Equivariant algebra

Genuine equivariance introduces transfer and norm operations and representation gradings that a Real descent theory must accommodate. With Chan, Mehrle, Sanchez Ocal, Osorno, Szczesny, and Verdugo, I study the realization of compatible transfer systems by pairings of N_∞-operads encoding equivariant distributive laws. We prove realization when the additive system is complete and for further families, and conjecture realization in general. In ongoing work with Noah Wisdom, I study representation-graded Mackey and Tambara functors. For C_2, we show that the additional automorphisms in a topological $\mathcal{RO}(C_2)$-grading of a Mackey functor are encoded by its canonical Burnside action, imposing no additional realization condition on grading symmetries.

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